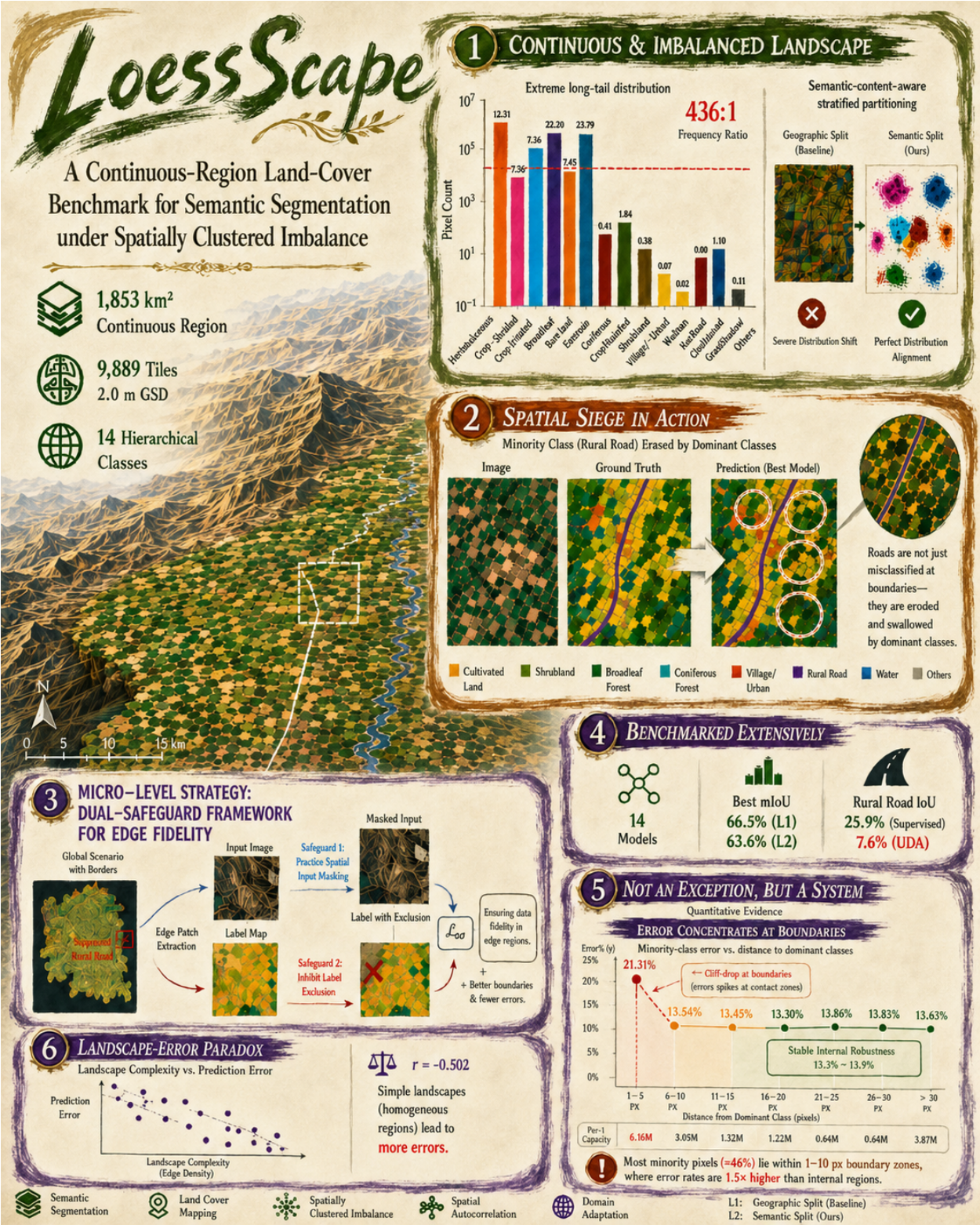


Graphical Abstract

LoessScape: A Continuous-Region Land-Cover Benchmark for Semantic Segmentation under Spatially Clustered Imbalance

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LoessScape: A Continuous-Region Land-Cover Benchmark for Semantic Segmentation under Spatially Clustered Imbalance

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Abstract

Existing remote-sensing semantic segmentation datasets are predominantly constructed under a scene-centric paradigm. By curating geographically isolated tiles, this paradigm disrupts spatial continuity and weakens the structural complexity of complete geographic entities. As a result, models evaluated on such idealized benchmarks often experience marked performance degradation when deployed in continuous real-world landscapes. To address this limitation, we introduce LoessScape, a semantic segmentation benchmark built entirely under a continuous-region survey paradigm. Instead of isolated sampling, the dataset preserves the complete spatial organization of a Loess Plateau region through a semantic-content-aware stratified partitioning strategy. This design explicitly exposes the challenge of Spatially Clustered Imbalance, in which geometrically thin minority classes are persistently embedded within strongly autocorrelated dominant terrains. Based on this dataset, we establish a comprehensive benchmark spanning hierarchical supervised segmentation on the L1 and L2 tasks and unsupervised domain adaptation under spatial distribution shift. Beyond aggregate empirical metrics, our cross-disciplinary diagnostic framework draws on landscape ecology to analyze the underlying failure patterns. Specifically, we identify a counterintuitive Landscape-Error Paradox and use spatial statistical analysis to characterize the phenomenon of Spatial Siege, showing that segmentation difficulty extends beyond localized boundary confusion into broader contextual erosion. Taken together, LoessScape provides a benchmark for studying spatial distribution shift and supports deployment-oriented land-cover mapping in continuous geographic environments. The dataset and project homepage are publicly available at <https://anyun-hy.github.io/LoessScape/>.

Keywords: Semantic Segmentation, Land Cover Mapping, Spatially Clustered Imbalance, Spatial Autocorrelation, Domain Adaptation

1. Introduction

Accurate and timely land cover mapping from high-resolution remote sensing imagery is fundamental to a myriad of critical applications, including sustainable urban planning, natural resource management, and environmental monitoring. The advent of deep learning, particularly semantic segmentation models, has revolutionized this field by enabling automated pixel-level classification with unprecedented precision. However, the performance and generalization capabilities of these data-intensive models are fundamentally underpinned by the quality and characteristics of the benchmark datasets used for their training and evaluation.

In recent years, the academic community has introduced a series of influential datasets that have significantly propelled algorithmic advancements. For instance, the ISPRS Potsdam dataset [1] spurred research in fine-grained urban classification, the DeepGlobe challenge [2] fostered broad community engagement, LoveDA [3] provided a crucial benchmark for domain

adaptation, and OpenEarthMap [4] pushed the frontier towards global generalization. However, a closer examination reveals a shared, fundamental limitation in their “scene-centric” construction paradigm. By assembling collections of representative yet geographically disjointed scenes, this paradigm does not fully preserve the spatial continuity and internal distribution structure of a complete geographic region [5]. This implicit filtering cultivates an idealized academic testing environment, systematically masking the systemic complexities of a continuous geographical entity. Consequently, state-of-the-art models often yield optimistic performance estimates in these curated settings, yet suffer significant performance degradation when deployed in real-world, large-scale mapping scenarios [3].

This scene-centric paradigm, while valuable for initial algorithmic validation, systematically sidesteps a more pervasive real-world problem: the endogenous class imbalance arising from the systemic complexity of a complete geographical entity. We define this challenge as **Spatially Clustered Imbalance**. Unlike the conventional long-tail problem, which is often described primarily in terms of sample frequency, this imbalance is further shaped by the spatial dependence emphasized by Tobler’s First Law of Geography [11]. As a result, land cover classes are not

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Table 1: A comprehensive comparison of LoessScape with representative high-resolution land cover datasets.

Dataset	Year	GSD (m)	Area (km ²)	Tiles	Classes	Primary Task	Paradigm	S.Cont.	H.Lab.	S.Coor.
ISPRS Vaihingen [1]	2013	0.09	1.38	33	6	Urban Classification	Curated urban scenes	×	×	✓
ISPRS Potsdam [1]	2013	0.05	3.42	38	6	Urban Classification	Curated urban scenes	×	×	✓
Zurich Summer [6]	2015	0.6	9.37	20	8	Urban Classification	Curated urban scenes	×	×	×
SpaceNet [7]	2018	~0.5	~2,544	~6,000	2	Object Detection	Curated urban scenes	×	×	✓
DeepGlobe [2]	2018	0.5	~1,717	1,146	7	Land Cover Mapping	Large-scale discrete	×	×	×
GID [8]	2020	4.0	~75,900	150	5	Land Cover Mapping	Large-scale discrete	×	×	✓
LandCoverNet [9]	2020	10.0	~30,000	~1,900	7	Land Cover Mapping	Large-scale discrete	×	×	✓
LoveDA [3]	2021	0.3	536	5,987	7	Domain Adaptation	Paired typical scenes	×	×	×
OpenEarthMap [4]	2023	0.25-0.5	799	5,000	8	Global Generalization	Large-scale discrete	×	×	✓
WHU-OPT-SAR [10]	2022	5.0	~50,000	100	7	Multi-modal Seg.	Large-scale discrete	×	×	✓
LoessScape (Ours)	2026	2.0	1,853	9,888	14 (H)	Land Cover Mapping	Continuous Region Survey	✓	✓	✓

(Note: H = Hierarchical, S.Cont. = Spatial Continuity, H.Lab. = Hierarchical Labels, S.Coor. = Spatial Coordinates.)

randomly distributed, but form continuous and structured spatial patterns in which vast, homogeneous areas of dominant classes such as farmland naturally surround small and spatially clustered instances of rare classes such as Rural Roads. Statistical analysis of the proposed dataset provides empirical support for this observation by revealing strong spatial dependence and pronounced concentration patterns across land cover categories, which are further investigated in the subsequent benchmark and diagnostic experiments. Consequently, state-of-the-art models that perform well on curated, geographically disjointed benchmarks often suffer significant performance degradation when deployed in such real-world landscapes, revealing a substantial gap between laboratory evaluation and large-scale application readiness.

To bridge this critical gap, we introduce **LoessScape**, a benchmark dataset for land cover semantic segmentation constructed under a continuous-region survey paradigm rather than conventional scene-based curation. The main contributions of this work are as follows:

1. **A complete-region benchmark for studying continuous-landscape segmentation.** LoessScape is constructed from a continuous geographic region of the Chinese Loess Plateau, preserving geographic continuity and the real-world distribution structure of land cover classes instead of relying on geographically disjoint scene selection. This design enables the explicit representation and systematic investigation of Spatially Clustered Imbalance in continuous geographic landscapes. To support representative and fair assessment under this setting, we further develop a semantic-content-aware stratified sampling strategy together with a dedicated evaluation protocol.
2. **A multi-scale benchmarking framework for deployment-oriented assessment.** The dataset is annotated with a dual-tier semantic hierarchy, including a 7-class macro-level taxonomy and a 14-class micro-level taxonomy, which supports model evaluation at different semantic granularities. On this basis, we benchmark representative state-of-the-art segmentation models across multiple experimental settings, including fine-grained mapping and unsupervised domain adaptation. In addition, the release of precise spatial coordinates facilitates future spatially grounded studies, such as multi-source remote sensing fusion and terrain-aware analysis, thereby

extending the value of the dataset beyond conventional computer vision benchmarks.

3. **A cross-disciplinary diagnostic framework for revealing model failure mechanisms.** Beyond global metrics such as mIoU, we propose a cross-disciplinary diagnostic framework that combines landscape ecology statistics, category-wise error analysis, and boundary-error profiling to investigate how spatial organization influences model behavior in continuous landscapes. By relating ecological structure to prediction errors, this framework provides empirical support for several nontrivial patterns that are difficult to capture through conventional benchmark metrics alone. In particular, our analysis reveals a counterintuitive **Landscape-Error Paradox** and suggests that severe rare-class degradation is closely linked to **Spatial Siege**, where minority categories are persistently surrounded and suppressed by strongly autocorrelated dominant landscapes. These findings offer a more interpretable view of model limitations in continuous real-world regions and provide useful guidance for deployment-oriented evaluation and method development.

2. Related Work

We review existing high-resolution semantic segmentation datasets through the lens of their construction paradigms (Table 1). Rather than comparing dataset scale or task definitions, this perspective clarifies how construction strategies fundamentally determine which forms of model generalization can be effectively evaluated.

Existing dataset paradigms inherently rely on scene-centric curation. First, the **curated urban scenes** paradigm (e.g., ISPRS Vaihingen/Potsdam [1]) drives fine-grained urban classification. However, evaluating on isolated tiles systematically removes macroscopic spatial context. Second, the **large-scale discrete sampling** paradigm (e.g., DeepGlobe [2], OpenEarthMap [4]) targets global diversity by assembling disjoint geographic snapshots, inherently breaking continuous spatial structures. Third, the **paired typical scenes** paradigm (e.g., LoveDA [3]) links categorized urban and rural tiles for domain adaptation, yet still curates discrete environments rather than preserving a complete geographic entity. Collectively, these scene-based sampling

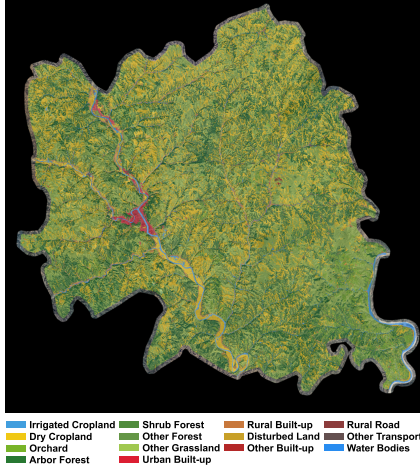


Figure 1: Macroscopic spatial organization of land cover across the continuous study region. The overlay demonstrates that different minority categories (e.g., human settlements) are closely associated with underlying terrain conditions, providing empirical evidence for the structural challenge of Spatially Clustered Imbalance.

methods provide limited support for confronting the continuous distribution patterns that emerge within a complete, real-world geographical system.

In contrast, LoessScape introduces a **continuous region survey** paradigm for deployment-oriented benchmarking. Foundational scene-centric benchmarks are crucial for confirming basic visual recognition capabilities. However, realistic deployment necessitates testing within uncured, continuous environments. As shown in Table 1, LoessScape is the first to simultaneously provide complete spatial continuity over a large area (1,853 km²), a hierarchical labeling system, and precise geo-referencing. This design provides empirical infrastructure to capture and study the structural spatial patterns that discrete sampling naturally masks.

This methodological evolution concretely exposes an application-level condition: Spatially Clustered Imbalance. Inside complete real-world landscapes, minority classes are rarely scattered randomly; instead, they are spatially aggregated and besieged by vast, homogeneous dominant terrains. By explicitly maintaining this natural spatial dependence, LoessScape serves as a more realistic testbed for models to confront real-world distributions and supports future research on large-scale deployment readiness.

3. LoessScape Dataset

3.1. Study Area: A Microcosm of Real-World Complexity

We construct LoessScape from a 1,852.76 km² continuous region within China’s Loess Plateau. This region was selected because its land cover configuration is strongly shaped by the interaction between geomorphology and long-term human activity. Recognized as a classic coupled human and natural system (CHANS) [5], the area is characterized by extensive loess tablelands and dense gully networks, which together impose substantial constraints on settlement patterns, agricultural use, and vegetation distribution.

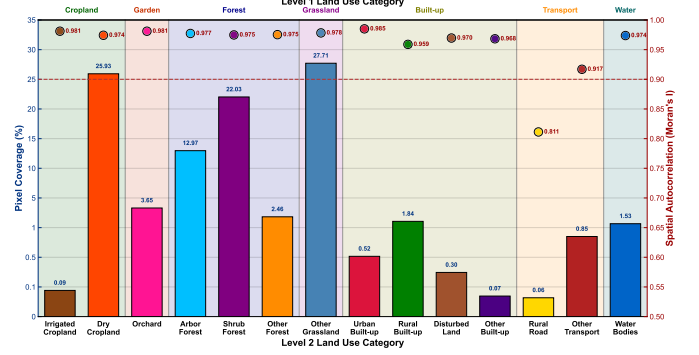


Figure 2: Quantitative distribution of the dual-granularity label space. The horizontal axis delineates the mapping between the 7 macro-level (L1) and 14 fine-grained (L2) categories. The left vertical axis (bars) shows the total pixel coverage depicting a pronounced long-tail distribution. The right vertical axis (scatter plot) overlays the Global Moran’s I values for each respective L2 class, illustrating the persistent spatial autocorrelation across all frequency tiers.

These geographic constraints give rise to a highly structured spatial organization of land cover, rather than a random mosaic of isolated categories. As illustrated in Figure 1, settlements are concentrated in comparatively accessible valley and tableland areas, whereas forests and grasslands are more closely associated with mountainous and gully-dissected terrain. For this reason, the study area provides a suitable setting for examining the challenge addressed in this work, namely Spatially Clustered Imbalance, in which minority classes are not only infrequent but also embedded within strongly autocorrelated dominant landscapes. By preserving this continuous regional structure, LoessScape enables evaluation under spatial conditions that are typically weakened or removed by geographically disjoint scene-based curation.

3.2. Hierarchical Classification and Long-Tail Distribution

To reflect the multi-scale requirements of real-world land cover mapping, LoessScape adopts a two-tier hierarchical classification system. The annotation schema comprises 7 macro-level categories at level 1 and 14 fine-grained categories at level 2. This dual-granularity label space supports evaluation at different semantic resolutions, covering both broad regional land cover characterization and more detailed environmental analysis.

More importantly, Figure 2 reveals two coupled statistical properties of the dataset. The descending bars on the primary axis show a pronounced long-tail distribution, with the pixel frequency ratio between the dominant class and the rarest class, Rural Roads, reaching approximately 436:1. At the same time, the overlaid scatter plot on the secondary axis shows that all categories maintain high Global Moran’s I [12] values ($I > 0.81$), including those with very limited pixel coverage.

This combination is important for understanding the difficulty of the dataset. The rarest classes are not approximately random or spatially scattered occurrences; instead, they remain strongly spatially organized despite their low overall frequency. In other words, minority classes in this dataset are both scarce in area and embedded in structured geographic contexts. This characteristic makes the class imbalance problem fundamentally different from

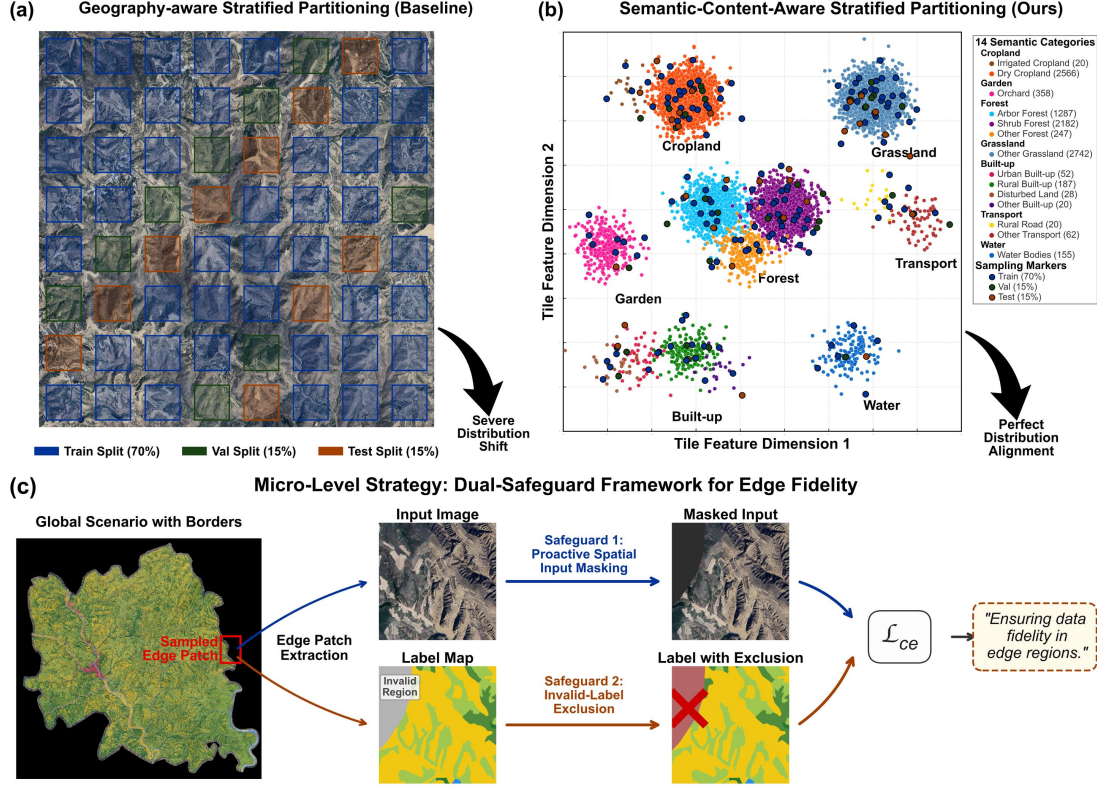


Figure 3: Methodological framework for data partitioning. (a) Conventional Geography-aware partitioning which induces severe distribution shift due to spatial autocorrelation. (b) The proposed Semantic-Content-Aware Stratified Partitioning, which achieves balanced alignment by sampling within the semantic label space. (c) Micro-level Dual-Safeguard Strategy for ensuring data fidelity in edge regions.

a purely frequency-based long-tail setting and directly motivates the partitioning design discussed in the following section.

3.3. Data Partitioning: A Validation-Driven Methodology

3.3.1. Methodological Framework: From Geographic Space to Semantic Content

For large, continuous geographical extents, the conventional baseline is **Geo-Partitioning**, which simply divides the region into spatial blocks according to geographic coordinates (Figure 3a). However, as established in the preceding analysis, this spatially intuitive strategy is not well suited to landscapes with strong spatial autocorrelation. Because rare categories are tightly clustered rather than uniformly distributed, geographic partitioning tends to separate minority classes unevenly across subsets, resulting in substantial distribution shifts that exacerbate imbalance and may bias downstream evaluation.

A common pragmatic remedy is to apply a **Manually Refined Partition**, which mitigates the most severe imbalances through human-guided tile adjustment. Although this refinement can reduce extreme outliers relative to the naive baseline, it remains a heuristic and non-scalable solution. Its effectiveness depends heavily on manual intervention, making it difficult to reproduce consistently in large-scale settings.

To obtain a more balanced and reproducible partition under these structural constraints, a shift in partitioning principle is required. We therefore adopt a content-aware methodology that transfers the basis of data splitting from geographic coordinates

to statistical semantic content. Driven by a **Semantic-Content-Aware Stratified Partitioning** strategy (Figure 3b), this framework yields the proposed **Tile-Stratified Partition** and follows a two-level design to mitigate the partition bias induced by spatial autocorrelation.

At the macro level, corresponding to inter-tile distribution, the Tile-Stratified Partition quantifies the pixel-level class abundance of each candidate tile and constructs the stratification basis entirely in the semantic label space. In this way, the representation of rare and highly clustered classes can be better preserved across the training, validation, and test subsets.

At the micro-level, this data-centric philosophy further extends to the integrity of individual samples. To address this issue, we adopt a Dual-Safeguard Strategy for handling Out-of-Distribution (OOD) edge regions (Figure 3c). By combining proactive spatial input masking with a secondary invalid-label exclusion mechanism during loss computation, this strategy prevents feature contamination from unannotated areas and improves data fidelity within the sampled subsets.

3.3.2. Quantitative Validation: A Multi-faceted Assessment

To empirically evaluate the effectiveness of the proposed partitioning strategy, we conduct a quantitative assessment from multiple complementary perspectives. Specifically, the proposed Tile-Stratified Partition is compared with two conventional baselines: Geo-Partitioning and the Manually Refined Partition. The assessment is organized across four dimensions,

covering macroscopic class proportions, cross-scale variance, information-theoretic distributional similarity, and overall quality assessment.

A. Macroscopic Class Proportion Consistency. Statistical consistency across data subsets is an important prerequisite for controlled evaluation, requiring the class distribution within each subset to remain close to the overall dataset distribution. Figure 4 examines this property at the pixel-percentage level. Compared with the two conventional baselines, the proposed Tile-Stratified Partition yields a more concentrated alignment of the training, validation, and test subsets with the global class proportions.

This advantage is particularly important for rare categories in the long-tail distribution, for which small absolute deviations may correspond to substantial relative shifts. For example, an increase from 0.01% to 0.04% represents a fourfold relative change, even though the absolute difference remains numerically small. By reducing such relative fluctuations across subsets, the proposed method improves pixel-level distribution consistency across both frequent and infrequent classes.

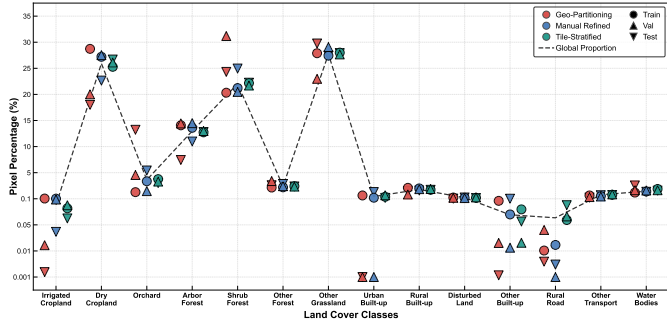


Figure 4: Pixel-percentage consistency across partitioning strategies. The proposed Tile-Stratified Partition (green) shows a closer alignment of the training, validation, and test subsets with the global dataset distribution (grey dashed line) than the two conventional baselines.

B. Cross-Scale Distributional Consistency. We further examine partition quality from both representational stability, reflected by class presence across subsets, and pixel-level abundance stability across spatial scales. Figure 5 uses the Coefficient of Variation (CV) to compare slice-level and pixel-level consistency. This distinction is important because balanced class occurrence across subsets does not necessarily imply balanced pixel-level abundance. The results show that the proposed Tile-Stratified Partition consistently yields lower Slice CV and Pixel CV across most classes than the conventional baselines. Notably, the gap between these two metrics is also reduced, indicating better agreement between class presence and actual pixel abundance across subsets. Figure 6 further supports this pattern, showing that more categories under the proposed strategy cluster near the low-CV diagonal region. Specifically, the proposed method achieves a consistency rate under the gap criterion of ≤ 0.10 of 50.0%, compared with 28.6% for the Manually Refined Partition and 7.1% for Geo-Partitioning.

C. Global Distributional Similarity. We further assess partition consistency from a global distributional perspective using the Kullback-Leibler (KL) divergence, which measures the overall shift between probability distributions. Figure 7 visu-

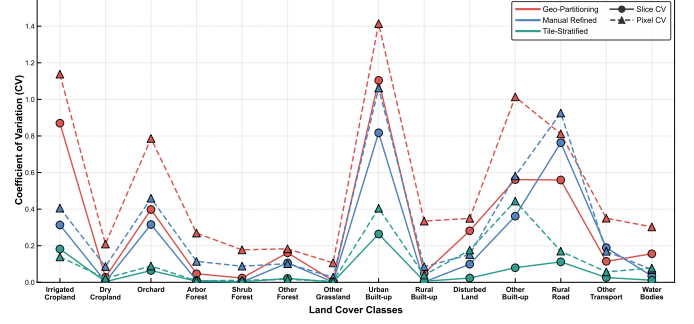


Figure 5: Cross-scale comparison of slice-level CV and pixel-level CV across land-cover classes under different partitioning strategies. The proposed Tile-Stratified Partition generally yields lower CV values and a smaller gap between the two metrics than the conventional baselines.

alizes the pairwise KL divergence among training, validation, and test subsets across different partitioning strategies. Compared with Geo-Partitioning and the Manually Refined Partition, the proposed Tile-Stratified Partition yields the lowest within-strategy KL values (≤ 0.0069) among its subsets. Meanwhile, the Manually Refined Partition partially moderates the distributional drifts associated with Geo-Partitioning but still exhibits noticeable subset-level anomalies. These results indicate that the proposed method preserves a higher degree of overall distributional similarity, providing a global corroboration of the earlier class-proportion and cross-scale consistency analyses.

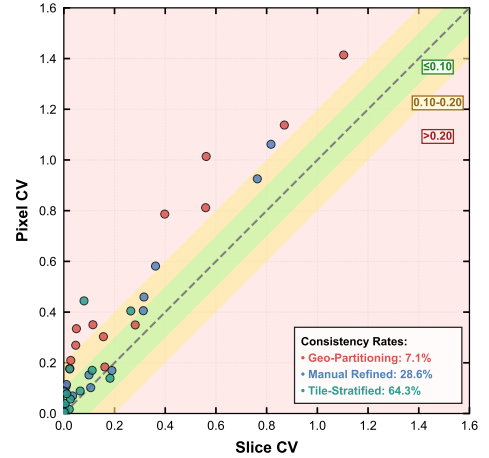


Figure 6: Scatter plot of pixel-level CV versus slice-level CV under different partitioning strategies. The proposed Tile-Stratified Partition places more classes within the high-consistency region with a gap of ≤ 0.10 , indicating closer agreement between slice-level and pixel-level distribution stability.

D. Cross-Metric Quality Synthesis. As a final integrated summary, Figure 8 provides a compact cross-metric comparison of the three tile partitioning strategies. Consistent with the results in Sections 3.3.2A–C, the proposed Tile-Stratified Partition, derived from our Semantic-Content-Aware Stratified Partitioning strategy, maintains the most favorable overall profile across the primary deviation and information-theoretic metrics, whereas Geo-Partitioning performs worst overall and the Manually Refined Partition provides only partial improvement. This indicates that the advantages of the proposed semantic-content-

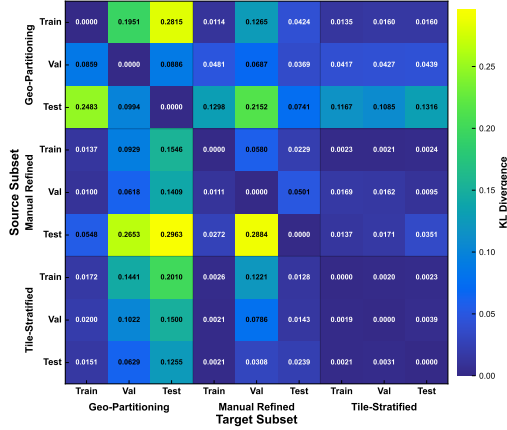


Figure 7: Heatmap of Kullback-Leibler (KL) divergence between subset pairs under different partitioning strategies. The proposed Tile-Stratified Partition yields the lowest within-strategy divergence values in its internal 3×3 block, indicating the highest degree of global distributional similarity among its training, validation, and test subsets.

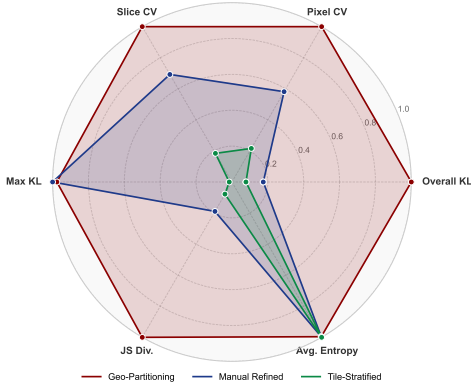


Figure 8: Radar chart synthesizing the evaluation of the three tile partitioning strategies across six statistical dimensions. The figure serves as a compact visual corroboration of the preceding analyses, demonstrating that the proposed Tile-Stratified Partition yields the most favorable overall profile across the primary deviation and information-theoretic metrics. (Note: Each metric is Min-Max normalized to a [0, 1] scale to facilitate qualitative visual comparison; lower values denote better stability for variance and divergence metrics, while higher values indicate better balance for entropy).

aware partitioning strategy are not confined to a single statistical property, but remain consistent across multiple aspects of tile-level distribution quality. The radar chart is therefore used here as a qualitative synthesis of the preceding quantitative analyses.

3.4. Dataset Overview and Composition

The finalized LoessScape dataset comprises 9,888 image tiles, each with an image size of 512×512 pixels. Based on the partitioning methodology described in Section 3.3, the data is divided into a training set (6,921 tiles, $\sim 70\%$), a validation set (1,477 tiles, $\sim 15\%$), and a test set (1,490 tiles, $\sim 15\%$).

A defining characteristic of this dataset is the presence of Spatially Clustered Imbalance. Unlike idealized assumptions of uniform class occurrence, the land-cover categories in this dataset exhibit a long-tail frequency distribution in which minority classes are geographically concentrated, as established in Section 3.2. This distributional structure reflects the inherent

complexity of real-world loess landscapes. The evaluation in Section 3.3.2 confirms that the proposed partitioning strategy preserves this structure across all subsets, thereby establishing a consistent foundation for assessing model performance under realistic imbalance conditions.

Furthermore, the spatial coordinates associated with every tile enable research beyond the immediate scope of this study. This spatial information provides a basis for integrated analyses and cross-modal investigations, such as correlating semantic categories with underlying geomorphological features through the fusion of labels with auxiliary data sources such as Digital Elevation Models (DEMs).

4. Experiments and Analysis

4.1. Implementation Details

Experimental Setup. All experiments were conducted on Ubuntu 22.04 LTS using PyTorch [13] 2.5 with CUDA 12.4, on a workstation equipped with four NVIDIA GeForce RTX 3090 Ti GPUs with 24 GB memory each. To improve reproducibility, all random seeds were fixed to 42. Synchronized Batch Normalization (SyncBN) was enabled across all GPUs to stabilize batch statistics during multi-GPU training.

Backbone Initialization. All networks were initialized with ImageNet-1K [14] pre-trained weights. For CNN-based models, we used official PyTorch checkpoints, with ResNet-50 [15] as the shared backbone for U-Net, FCN, DeepLabV3+, PSP-Net, and other ResNet-50-based CNN baselines, while HRNet was initialized with HRNet-W48 weights. For Transformer-based models, SegFormer was initialized with MiT-B2/MiT-B5 weights released by the original authors, while SETR used ViT-Large [16] pre-trained on ImageNet-21K.

Optimization Strategy. We adopted architecture-specific optimization settings following common practice for different model families. For CNN-based models, we used SGD with an initial learning rate of 0.01, momentum of 0.9, and weight decay of 5×10^{-4} . For Transformer-based models, we used AdamW with an initial learning rate of 6×10^{-5} , betas of (0.9, 0.999), and weight decay of 0.01. A $10\times$ learning-rate multiplier was applied to the decoder head to accelerate task adaptation while retaining the pre-trained backbone initialization. All models were trained for 100 epochs using a poly learning-rate schedule with power = 0.9 and a linear warmup of 1,500 iterations.

Data Configuration. Input images were resized to 512×512 pixels. The effective batch size was set to 32 for most models, corresponding to 8 samples per GPU, and reduced to 16 for the memory-intensive SETR with ViT-Large backbone. Standard data augmentation included random horizontal and vertical flips, as well as color jittering with brightness, contrast, and saturation set to 0.2 and hue set to 0.05. Inputs were normalized using the standard ImageNet mean and standard deviation.

Handling Class Imbalance. For the 14-class task, we used a class-balanced cross-entropy loss. Class weights were computed using a square-root variant of median frequency balancing, $w_c = \sqrt{\text{median}(f)/f_c}$, where f_c denotes the pixel frequency of class c . To avoid excessively large gradients from extremely rare classes

Table 2: Performance of baseline models on the L1 (7-class) task (%).

Method	Backbone	Cropland	Garden	Forest	Grassland	Built-up	Transport	Water	mIoU	OA	F1
FCN [17]	ResNet-50	65.97	51.80	65.25	56.19	57.62	46.62	67.19	58.66	76.66	73.68
U-Net [18]	ResNet-50	65.38	48.21	64.24	54.35	55.92	44.23	64.69	56.72	75.70	72.06
DeepLabV3+ [19]	ResNet-50	66.68	53.22	66.72	57.89	57.59	46.72	68.00	59.54	77.64	74.37
PSPNet [20]	ResNet-50	64.47	50.33	63.81	54.76	55.01	43.01	62.41	56.26	75.47	71.72
UperNet [21]	ResNet-50	63.53	48.75	63.38	53.20	54.01	43.88	63.47	55.74	74.79	71.29
LinkNet [22]	ResNet-50	62.99	45.91	62.57	52.30	53.78	40.24	62.28	54.30	74.12	70.00
FPN [23]	ResNet-50	62.51	43.89	62.06	51.49	53.18	40.33	61.28	53.54	73.63	69.34
U-Net++ [24]	ResNet-50	62.85	44.65	62.16	51.35	53.70	40.58	60.07	53.62	73.73	69.44
PAN [25]	ResNet-50	57.05	37.76	56.13	44.58	49.08	33.46	53.39	47.35	68.54	63.80
MANet [26]	ResNet-50	57.80	39.26	57.47	45.93	48.67	35.61	55.00	48.53	69.55	64.93
HRNet [27]	W48	66.02	52.22	65.31	55.93	56.20	47.24	66.63	58.51	76.66	73.57
SegFormer [28]	MiT-B2	65.62	51.12	65.09	55.28	58.33	48.00	66.65	58.59	76.36	73.64
SegFormer [28]	MiT-B5	69.96	58.85	69.64	61.52	62.18	53.62	72.11	63.98	80.11	77.85
SETR [29]	ViT-L	72.01	62.17	71.79	65.73	65.00	53.90	75.11	66.53	82.05	79.70

Note: Models are grouped by backbone family, with ResNet-based models listed above the horizontal separator. Bold indicates the overall best result in each column, while bold-italic highlights the best performance specifically within the ResNet-based group.

such as Rural Roads, the weights were clipped to the range [0.5, 5.0]. This design is intended to increase the contribution of tail classes while limiting numerical instability caused by overly large class weights.

4.2. Experiment 1: Supervised L1 Benchmark

To establish a supervised baseline for LoessScape, we benchmark a set of widely used semantic segmentation models on the L1 (7-class) task. This coarse-grained setting serves two purposes. First, it provides a reference for assessing the overall difficulty of the dataset under standard supervised learning. Second, it examines whether the relative performance ordering across major model families is consistent with commonly observed trends in semantic segmentation.

Model Selection. We evaluate 14 architectures spanning three representative model families: 1) CNN-based models, including FCN, U-Net, DeepLabV3+, PSPNet, UperNet, LinkNet, FPN, U-Net++, PAN, and MANet; 2) high-resolution architectures, represented by HRNet-W48; and 3) Transformer-based models, including SETR with ViT-L and SegFormer with MiT-B5 and MiT-B2 backbones. All models are trained and evaluated under the unified protocol described in Section 4.1 to maintain a controlled comparison across architectures.

Results and Discussion. Table 2 shows a clear and interpretable performance hierarchy across model families.

Overall performance. Transformer-based models achieve the highest overall mIoU on the L1 task. SETR obtains the highest mIoU at 66.53%, followed by SegFormer MiT-B5 at 63.98%. Among the non-Transformer models, DeepLabV3+ is the most competitive (59.54%), followed closely by FCN (58.66%), SegFormer MiT-B2 (58.59%), and HRNet-W48 (58.51%), but their performance remains below that of the top Transformer models. This pattern suggests that global context modeling may be beneficial for the continuous and spatially heterogeneous landscapes covered by LoessScape, where classes often extend over large areas while remaining fragmented by terrain and landform variation.

Class-wise behavior. Performance varies substantially across classes. Water and Cropland achieve relatively high IoU values for most models, whereas Transport consistently has the lowest IoU across models. This pattern is consistent with the class distribution described in Section 3.2, where Transport accounts for only a small proportion of total pixels. In addition to limited training supervision, Transport categories in this dataset are often expressed as thin and elongated structures such as Rural Roads. Under this geometric characteristic, even small localization deviations can produce broken predictions and noticeable boundary loss. These class-specific failure patterns are examined further in later analyses.

Dataset difficulty. The overall performance range indicates that the L1 task is non-trivial even under coarse-grained labeling. Across the 14 evaluated models, mIoU ranges from 47.35% to 66.53%, showing that substantial performance differences remain even at the coarse-grained L1 level. This difficulty is consistent with the geographic properties of the study region. Dense gully networks, dissected terrain, strong scale variation among land-cover objects, and illumination changes induced by slope and aspect all increase boundary ambiguity and reduce intra-class consistency. As a result, LoessScape remains challenging even before moving to the finer-grained L2 setting.

Overall, the L1 benchmark provides a useful baseline for the subsequent experiments. It shows that LoessScape supports interpretable cross-architecture comparison under a standard supervised setting, while also presenting sufficient difficulty to distinguish model capabilities on a continuous real-world landscape.

4.3. Experiment 2: Supervised L2 Benchmark

To further assess model behavior under finer semantic granularity, we evaluate the same 14 architectures on the L2 (14-class) task. This experiment serves two purposes. First, it examines whether the relative performance ordering observed on the L1 task remains stable after the label space is expanded from 7 parent classes to 14 child classes. Second, it quantifies the extent

Table 3: Performance of baseline models on the L2 (14-class) task (%).

Method	Backbone	Irrig	Dry	Orch	Arbor	Shrub	OthFor	Grass	Urban	Rural	Dist	OthBt	Road	OthTr	Water	mIoU	OA	F1
FCN	ResNet-50	74.0	65.3	52.0	50.3	54.1	41.1	55.9	73.2	56.1	41.8	51.0	22.0	53.2	69.0	54.21	72.43	69.26
U-Net	ResNet-50	72.5	63.0	47.9	46.9	52.0	38.4	53.0	71.6	53.1	40.9	42.2	21.3	50.2	66.1	51.36	70.24	66.78
PSPNet	ResNet-50	72.9	64.2	52.5	49.4	53.6	38.4	56.0	77.5	53.5	41.6	48.9	18.1	50.0	66.7	53.09	71.93	68.10
LinkNet	ResNet-50	69.8	60.8	44.6	44.8	49.2	37.1	50.6	65.6	50.5	36.7	39.0	17.7	45.9	60.4	48.05	68.17	63.79
FPN	ResNet-50	68.1	61.5	45.5	45.6	50.5	36.3	51.5	68.2	51.4	35.3	38.8	18.6	46.9	63.7	48.72	68.92	64.36
DeepLabV3+	ResNet-50	73.9	66.7	54.9	51.2	56.3	41.7	58.0	76.6	57.0	43.0	48.1	22.6	54.1	69.7	55.28	73.84	70.12
UperNet	ResNet-50	70.2	62.9	50.0	46.3	52.0	36.9	52.9	74.6	52.9	40.9	41.8	20.6	50.7	65.8	51.31	70.16	66.69
U-Net++	ResNet-50	69.5	61.2	45.6	44.1	49.5	35.2	50.5	65.2	51.0	36.1	38.7	19.0	47.0	61.1	48.13	68.15	63.88
PAN	ResNet-50	65.2	56.8	40.4	37.9	44.7	25.9	45.6	69.4	46.8	28.8	36.3	11.7	41.0	55.2	43.27	63.48	58.82
MANet	ResNet-50	61.5	55.0	38.3	34.6	41.0	19.7	43.2	68.4	45.4	28.2	30.7	5.2	38.9	52.6	40.19	60.83	55.34
HRNet	W48	72.7	64.3	52.2	49.6	54.4	40.0	56.0	73.7	55.3	37.9	46.6	24.9	54.3	67.9	53.55	72.25	68.75
SegFormer	MiT-B2	77.2	64.5	52.1	49.0	53.2	36.3	55.0	75.3	55.9	48.0	52.6	22.6	53.7	68.1	54.53	71.69	69.48
SegFormer	MiT-B5	78.4	69.0	58.1	54.9	58.9	45.3	60.6	80.4	59.9	52.3	53.3	27.6	58.4	73.5	59.32	75.97	73.56
SETR	ViT-L	77.7	72.7	64.4	62.5	65.0	53.8	67.5	83.0	63.0	59.6	59.8	25.9	59.0	76.5	63.60	80.33	76.85

Note: Models are grouped by backbone family, with ResNet-based models listed above the horizontal separator. Bold indicates the overall best result in each column. Column abbreviations: Irrig=Irrigated Cropland, Dry=Dry Cropland, Orch=Orchard, Arbor=Arbor Forest, Shrub=Shrub Forest, OthFor=Other Forest, Grass=Grassland, Urban=Urban Built-up, Rural=Rural Built-up, Dist=Disturbed Land, OthBt=Other Built-up, Road=Rural Road, OthTr=Other Transport, Water=Water Bodies.

Table 4: Head-Tail performance disparity on the L2 (14-class) task (%). Bold indicates the best value in each column. For Gap, lower is better.

Method	Backbone	mIoU	mIoU-Head	mIoU-Tail	Gap
FCN	ResNet-50	54.21	56.08	52.33	3.75
U-Net	ResNet-50	51.36	53.38	49.34	4.04
PSPNet	ResNet-50	53.09	55.28	50.90	4.39
LinkNet	ResNet-50	48.05	51.00	45.10	5.90
FPN	ResNet-50	48.72	51.30	46.14	5.16
DeepLabV3+	ResNet-50	55.28	57.52	53.03	4.49
UperNet	ResNet-50	51.31	53.02	49.61	3.41
U-Net++	ResNet-50	48.13	50.81	45.45	5.37
PAN	ResNet-50	43.27	45.22	41.33	3.89
HRNet	W48	53.55	55.59	51.51	4.08
MANet	ResNet-50	40.19	41.89	38.49	3.40
SegFormer	MiT-B2	54.53	55.33	53.74	1.59
SegFormer	MiT-B5	59.32	60.73	57.91	2.82
SETR	ViT-L	63.60	66.23	60.97	5.26

Note: Average Gap: 4.11%. Largest Gap: 5.90% (LinkNet). Smallest Gap: 1.59% (SegFormer MiT-B2).

of performance degradation from L1 to L2 and analyzes whether this degradation is distributed evenly across classes.

Model Selection. We use the same model configurations and training protocols as in Section 4.1 to maintain a controlled comparison across granularity levels. The only change lies in the label space: the original 7 parent classes are subdivided into 14 child classes according to the hierarchical taxonomy defined in Section 3.2.

To further examine class-wise performance disparity, we divide the 14 L2 classes into head and tail groups according to their L1 parent categories. In LoessScape, the seven L1 categories fall into two natural groups by training-set pixel proportion: four dominant categories (Cropland, Garden, Forest, and Grassland; each > 3%) and three minority categories (Built-up, Transport, and Water Bodies; each < 3%). Accordingly, the head group consists of the 7 L2 children of the four dominant L1 categories, and the tail group consists of the 7 L2 children of the three minority L1 categories. We then define mIoU-Head and mIoU-Tail as the

mean IoU computed within each group, and Gap = mIoU-Head – mIoU-Tail to quantify the head-to-tail performance disparity. Table 4 reports these values for each model.

Results and Discussion. The L2 benchmark preserves the same top-2 models observed on the L1 task, with SETR and SegFormer MiT-B5 achieving the highest mIoU values in both settings. At the same time, all models show lower mIoU on L2 than on L1, indicating that the finer label space introduces additional difficulty across architectures.

Overall performance drop. Comparing Tables 2 and 3, we observe a consistent decline in mIoU when moving from L1 to L2. SETR, the best-performing model in both settings, decreases from 66.53% on L1 to 63.60% on L2, corresponding to a drop of 2.93 percentage points. Several CNN-based baselines show larger degradation. For example, MANet decreases from 48.53% to 40.19%, a drop of 8.34 points. These results indicate that fine-grained labels place higher demands on feature discrimination, boundary localization, and context aggregation than the coarser L1 setting.

Head-tail disparity. Table 4 shows that head classes generally achieve higher mIoU than tail classes. Averaged over all models, mIoU-Head exceeds mIoU-Tail by 4.11 percentage points. The largest gap is observed for LinkNet at 5.90%, whereas the smallest gap is observed for SegFormer MiT-B2 at 1.59%. This gap, however, should be interpreted together with the absolute mIoU values. For SegFormer MiT-B2, the small gap is accompanied by relatively similar head and tail performance, rather than uniformly low scores. By contrast, MANet also shows a relatively small gap at 3.40%, but this reflects low mIoU on both head and tail classes. Therefore, a smaller Gap does not by itself imply stronger performance on tail categories.

Tail-class heterogeneity. Performance within the tail group varies substantially across classes. Water achieves relatively high IoU values for many models, reaching 76.45% for SETR, whereas Rural Road remains the weakest category overall. The best result on Rural Road is 27.6%, obtained by SegFormer MiT-B5, while MANet yields only 5.2%. This contrast indicates that class frequency alone is insufficient to explain the variation in L2 performance. Differences in geometric structure and local

Table 5: UDA performance comparison on the L2 (14-class) task (%). Bold indicates the best per-class IoU and summary metrics among UDA methods; Source Only and Oracle are excluded from bolding. OA and Kappa are reported for completeness. All methods use SegFormer MiT-B5 backbone.

Method	Type	Irrig	Dry	Orch	Arbor	Shrub	OthFor	Grass	Urban	Rural	Dist	OthBt	Road	OthTr	Water	mIoU	mF1	OA	Kappa
Source Only	Base	71.45	54.94	4.00	29.23	36.40	17.81	39.82	73.87	47.60	36.24	29.55	6.49	43.99	48.85	38.59	52.50	57.50	0.4501
CBST [30]	ST	73.16	54.24	3.55	29.28	34.76	19.08	39.08	75.63	46.77	32.63	27.08	6.66	44.42	49.09	38.25	52.03	56.94	0.4412
AdaptSeg [31]	Adv	75.64	55.65	3.76	29.88	34.64	19.16	41.20	78.84	47.66	34.33	27.49	8.06	45.46	49.84	39.40	53.13	57.58	0.4535
DAFormer [32]	Tr+ST	75.56	51.29	4.56	29.92	36.11	18.99	38.77	79.45	48.89	38.02	33.55	7.14	46.51	46.33	39.65	53.51	57.22	0.4419
AdveNT [33]	Adv+Ent	75.04	55.85	3.71	29.73	34.89	19.35	40.37	78.02	47.70	33.70	27.66	8.24	45.59	50.77	39.33	53.09	57.42	0.4514
DACS [34]	Mix+ST	71.67	54.34	4.62	26.91	36.15	19.31	40.94	82.06	48.34	38.23	28.96	7.57	46.91	54.10	40.01	53.74	57.93	0.4513
Oracle	Sup	56.64	60.89	50.91	41.05	47.59	26.16	48.13	0.00	46.22	16.93	34.66	8.19	40.69	55.19	38.09	52.31	66.21	0.5671

Note: Base: no adaptation (lower bound). Sup: fully supervised on target domain (upper bound). ST: Self-Training. Adv: Adversarial alignment. Ent: Entropy minimization. Mix: Cross-domain mixed sampling. Tr: Transformer-based architecture.

context may also contribute. In particular, thin and elongated objects such as Rural Roads are more susceptible to boundary erosion and fragmented predictions, especially when they are adjacent to dominant agricultural classes.

Implications. The L2 benchmark shows that fine-grained classification in the Loess Plateau introduces challenges beyond an increase in the number of categories. The observed degradation is not evenly distributed across classes, and the weakest performance is concentrated in several rare and spatially constrained categories. This pattern is consistent with the spatial analysis in Section 3.2, where Rural Roads exhibit both the lowest frequency and strong spatial autocorrelation with Moran’s $I \approx 0.81$. Taken together, these results suggest that segmentation difficulty at the L2 level is associated not only with class frequency, but also with structured spatial context. This observation provides motivation for the subsequent domain adaptation experiments.

4.4. Experiment 3: Unsupervised L2 UDA Benchmark

We benchmark UDA methods on the L2 (14-class) task to assess whether knowledge learned from spatially distinct labeled source regions can be transferred to unlabeled target regions under spatial distribution shift.

4.4.1. UDA Benchmark Setup and Evaluation

Experimental Design. We define source and target domains based on spatial partitioning rather than class distribution. The training set is divided into a 5×5 grid based on geographic coordinates. The source domain consists of tiles from the left two columns, accounting for approximately 40% of the training tiles by count, for which ground-truth labels are available during training. The target domain consists of tiles from the remaining three columns, accounting for approximately 60%, which are unlabeled during training. Pseudo-labels are generated from model predictions on these unlabeled target images. All methods are evaluated on the same held-out test split. The test split is fixed and independent of the source/target partition; it contains tiles from both the source and target regions by design.

Methods. We use SegFormer MiT-B5 as the backbone for all methods for consistency. We evaluate five representative UDA methods: (1) **CBST** (class-balanced self-training) [30], (2) **AdaptSeg** (adversarial output-space adaptation) [31], (3) **AdveNT** (adversarial training with entropy minimization) [33], (4) **DAFormer** (Transformer-based with self-training) [32], and

(5) **DACS** (domain adaptation via cross-domain mixed sampling) [34]. We also report **Source Only** (trained on the source domain only, without adaptation) as the lower-bound baseline and **Oracle** (Target-Supervised), i.e., supervised training on the target split with full labels, as a supervised target-domain reference under this spatial partition.

UDA effectiveness. Table 5 shows that four out of five UDA methods improve over Source Only (38.59%). DACS achieves the highest mIoU (40.01%) among the evaluated UDA methods, with a gain of 1.42 percentage points. CBST (38.25%) is the sole exception, falling marginally below Source Only, likely due to unreliable pseudo-label generation under spatial class imbalance. Although the overall gains remain limited in magnitude, positive improvements are observed for most methods.

Tail-class limitations. Despite these overall gains, improvement on tail classes remains limited. Although AdveNT achieves the highest Road IoU among UDA methods at 8.24%, marginally above even the Oracle reference, the best overall method DACS reaches only 7.57%, a gain of merely 1.08 points over Source Only (6.49%). This near-parity with Oracle likely reflects training variance on extremely rare classes rather than genuine superiority over supervised learning. These limited gains indicate that pseudo-label-based adaptation remains weak for spatially constrained linear structures under the present setting. One possible reason is that early pseudo-labels on thin structures are less reliable, which may limit the effectiveness of self-training. This observation is consistent with the spatial autocorrelation analysis in Section 3.2, where Rural Roads exhibit both strong spatial clustering and very low pixel frequency.

Oracle interpretation. Oracle yields 0.00% IoU on Urban Built-up because urban areas are geographically concentrated in the source region, while the target-domain training split contains no urban pixels. Beyond Urban Built-up, several other classes also exhibit reduced Oracle performance due to limited target-domain coverage. These patterns reflect the class-presence shift induced by the spatial partition, rather than a random split. Notably, Oracle achieves an overall mIoU of 38.09%, which falls below several UDA methods including DACS (40.01%), despite showing the highest OA (66.21%) and Kappa (0.5671) among all methods. This discrepancy arises because mIoU equally weights all 14 classes, so the zero contribution from Urban Built-up substantially penalizes the class-averaged metric even when prediction quality on the remaining classes is high. Therefore, Oracle should be interpreted as a target-domain reference under the present partition, rather than a universal upper bound.

Table 6: Ablation study of DACS on the L2 (14-class) task (%). DACS (full) corresponds to the DACS result reported in Table 5.

Variant	Ablation	Irrig	Dry	Orch	Arbor	Shrub	OthFor	Grass	Urban	Rural	Dist	OthBt	Road	OthTr	Water	mIoU	mF1	OA	Kappa
Fixed Thr.	-Pct	72.38	54.21	3.63	28.75	33.50	18.18	40.57	77.05	47.97	35.51	28.10	6.93	45.24	49.52	38.68	52.45	56.93	0.4410
Single Head	-DH	72.78	53.94	4.44	28.88	36.43	19.19	39.05	81.49	48.09	38.38	34.40	6.06	45.71	49.99	39.92	53.69	57.31	0.4457
Full	DH+Pct	71.67	54.34	4.62	26.91	36.15	19.31	40.94	82.06	48.34	38.23	28.96	7.57	46.91	54.10	40.01	53.74	57.93	0.4513
Full + CW	+CBL	74.03	54.26	4.68	30.08	34.75	20.23	33.63	80.71	48.44	39.28	33.21	13.61	48.49	60.14	41.11	55.10	55.26	0.4252

Note: Pct: per-class percentile thresholding for pseudo-labels. DH: dual-head architecture with joint L1 and L2 supervision. CBL: Class-Balanced Loss weighting. -: component removed from full model. +: component added to full model.

4.4.2. Ablation and Parameter Analysis

To further examine the best-performing UDA setting, we conduct ablation experiments on DACS and analyze the sensitivity of the Rare Class Sampling (RCS) parameters.

Ablation Study. We consider three design choices in DACS: (1) per-class percentile thresholding for pseudo-labels, (2) dual-head architecture with joint L1 and L2 supervision, and (3) class-balanced loss weights.

Ablation results. Table 6 summarizes the contribution of each component. Per-class percentile thresholding improves mIoU by 1.33 points (40.01 vs 38.68). The dual-head architecture yields a marginal improvement of 0.09 points (40.01 vs 39.92), suggesting that the L1 auxiliary supervision provides limited additional benefit under the new dataset version. Adding class-balanced loss further increases mIoU by 1.10 points (41.11 vs 40.01), with the most notable gain on Rural Roads (+6.04 points, from 7.57% to 13.61%). Unlike in the previous dataset version, Other Forest IoU does not decline under class-balanced loss (19.31% vs 20.23%), indicating that the re-weighting effect is more stable in the new annotation. OA and Kappa are more strongly influenced by frequent classes, so improvements on tail classes may not necessarily increase these summary metrics.

RCS parameter tuning. Given the large disparity in class-wise pixel counts, we further re-tune the Rare Class Sampling (RCS) parameters. Figure 9 shows that $min_pixels = 200$ and $\tau = 0.05$ achieve the highest mIoU among the tested settings.

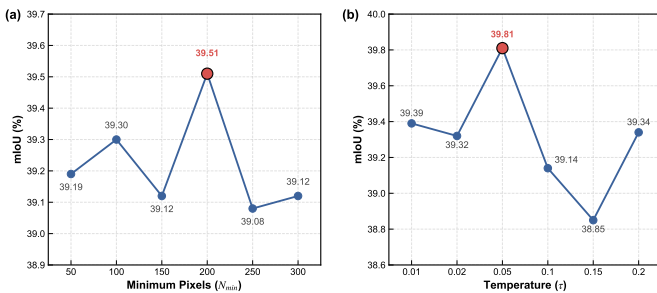


Figure 9: RCS parameter ablation. Left: min_pixels sweep with $\tau = 0.05$. Right: temperature sweep with $min_pixels = 200$.

Implications. UDA yields modest but consistent gains under the present spatial split. However, tail classes such as Rural Roads remain below 10% IoU across all evaluated methods, with even the best per-class result (AdveNT, 8.24%) falling far short of practical deployment thresholds. This indicates that structured spatial imbalance remains difficult to address under the present setting. These observations motivate the qualitative analysis in the next section.

4.5. Experiment 4: Qualitative Analysis of Segmentation Failures

The preceding experiments have revealed performance patterns consistent with the challenge of Spatially Clustered Imbalance through aggregate metrics. This section further examines model predictions to analyze characteristic failure modes, complemented by spatial analyses based on the geographic coordinates of the dataset.

4.5.1. Visual Comparison of Model Predictions

To systematically analyze segmentation quality, we select representative scenes containing diverse land cover configurations, with particular attention to regions where minority classes are embedded within dominant landscapes. Figure 10 presents predictions from supervised learning models on the L2 task, while Figure 11 shows results from UDA methods.

Among the supervised models, consistent differences emerge in the preservation of minority-class structures. FCN and U-Net exhibit pronounced fragmentation on Rural Roads, yielding discontinuous predictions in which linear segments are misclassified as the surrounding Shrub Forest class. Small isolated patches, such as Orchards and Water Bodies, are also less consistently preserved. DeepLabV3+ and PSPNet improve regional connectivity but produce boundary inconsistencies. HR-Net shows relatively better preservation of small objects. SETR preserves structural continuity more consistently: roads maintain better connectedness, and the model recovers scattered small objects that are often omitted by CNN architectures. However, SETR predictions show slightly coarser local transitions in some curved segments. SegFormer provides a comparatively balanced result, combining structural connectivity with smoother boundaries.

The domain adaptation comparison (Figure 11) shows increased prediction fragmentation under domain shift. Source Only predictions exhibit pixel-level noise, with narrow rivers and roads fragmented into isolated components or misclassified as surrounding background classes. AdaptSeg and AdveNT reduce this noise and preserve coarse regional structures, but fine-grained geometric details remain weakly recovered. DACS yields contiguous intra-class regions but exhibits boundary erosion on minority classes. DAFormer shows higher region consistency, sharper boundaries, and better preservation of the topological structure of linear features, particularly in scenes with complex elongated structures.

Figures 12 and 13 show row-normalized confusion matrices, in which each row represents the recall distribution of a true class over predicted labels. This quantitative analysis is consistent with the visual observations. For SETR, the model with the

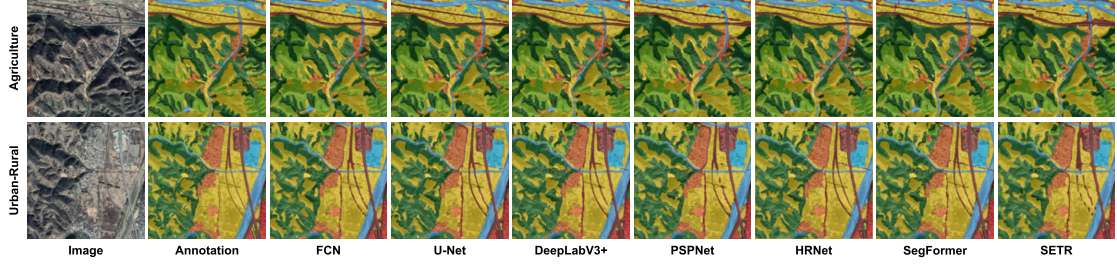


Figure 10: Visual comparison of supervised learning models on the L2 task. From left to right: FCN, U-Net, DeepLabV3+, PSPNet, HRNet, SegFormer MiT-B5, and SETR.

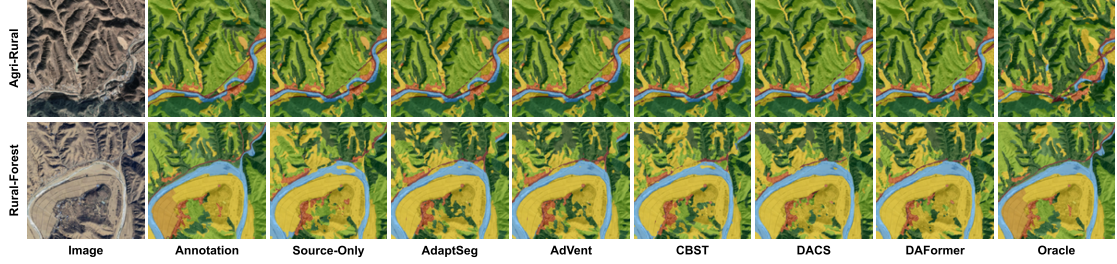


Figure 11: Visual comparison of UDA methods. From left to right: Source Only, AdaptSeg, AdveNT, CBST, DACS, DAFormer, and Oracle.

highest mIoU among the supervised set on the L2 task (Table 3), 58.9% of Rural Road pixels are correctly classified, while 3.1% are misclassified as Shrub Forest and 10.8% as Dry Cropland. These misclassification targets correspond to the classes that most frequently surround roads in the study area. This pattern aligns with the proposed Spatial Siege effect, in which minority classes are eroded by spatially adjacent dominant neighbors.

This misclassification pattern becomes more pronounced under domain shift. For DACS, the method with the highest mIoU among the UDA baselines, Rural Road recall decreases from 58.9% under SETR supervision to 8.2% on the same held-out test split, with 24.8% of pixels misclassified as Shrub Forest and 31.2% as Dry Cropland. This contrast indicates the vulnerability of linear tail classes under domain shift. It also suggests that pseudo-label-based adaptation remains difficult for extremely rare classes, especially when they are embedded in homogeneous majority regions.

4.5.2. Landscape Complexity and Prediction Error

To investigate the relationship between macroscopic landscape heterogeneity and model performance, we compute three landscape metrics from the ground-truth mask of each 512×512 test tile: Patch Density, defined as the total number of connected components (4-connectivity) across all classes divided by the tile area; Edge Density, the proportion of pixels whose 4-neighborhood contains at least one pixel of a different class; and the Shannon Diversity Index, $H = -\sum_{i=1}^m p_i \log_2 p_i$, where p_i is the pixel proportion of class i among the m classes present. Higher values of these metrics indicate more fragmented, multi-class landscapes. Each metric is then correlated with tile-level prediction error using Pearson’s linear correlation coefficient. To reduce sensitivity to the sparsity of rare classes at the tile level, we evaluate both class-averaged error ($1 - \text{mIoU}$) and frequency-weighted error ($1 - \text{OA}$). Figure 14 presents the re-

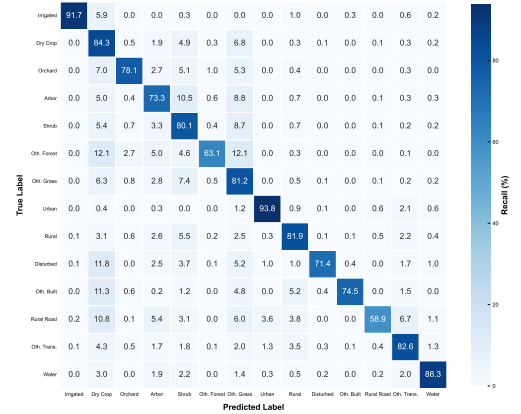


Figure 12: Row-normalized confusion matrix for SETR on the L2 task. Rural Road shows 58.9% correct classification, with 3.1% misclassified as Shrub Forest and 10.8% as Dry Cropland.

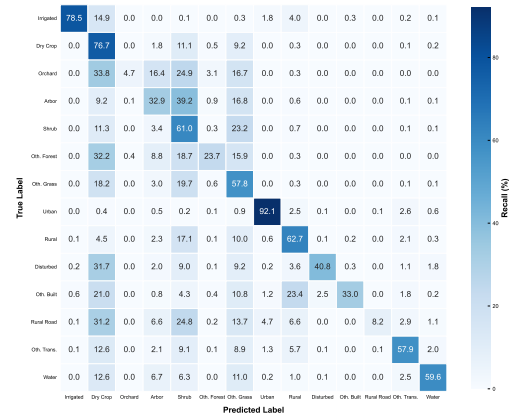


Figure 13: Row-normalized confusion matrix for DACS on the L2 task. Rural Road recall drops to 8.2%, with 24.8% misclassified as Shrub Forest and 31.2% as Dry Cropland under domain shift.

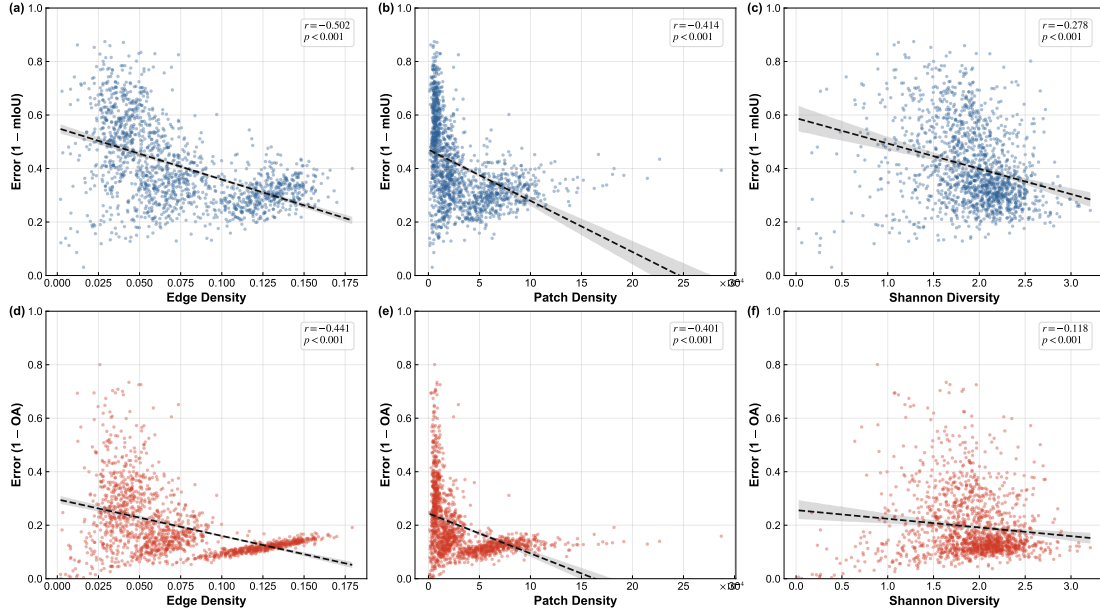


Figure 14: Correlation between landscape complexity metrics and tile-level prediction error. Edge Density, Patch Density, and Shannon Diversity all show statistically significant negative correlations with both mIoU error (top row) and accuracy error (bottom row). This pattern suggests that prediction errors are more concentrated in relatively homogeneous, majority-dominated landscapes.

sulting correlations across 1,490 test tiles.

All three landscape complexity metrics show statistically significant negative correlations with prediction error. As landscape complexity increases, tile-level error tends to decrease. Among the three metrics, Edge Density shows the strongest association, with $r = -0.502$ for $(1 - \text{mIoU})$ and $r = -0.441$ for $(1 - \text{OA})$, both with $p < 0.001$. Similar negative relationships are observed for Patch Density ($r = -0.414, -0.401$) and Shannon Diversity ($r = -0.278, -0.118$), although the association is weaker for Shannon Diversity, especially with frequency-weighted error.

This pattern is consistent with the Spatially Clustered Imbalance hypothesis. Low-complexity tiles often correspond to relatively homogeneous landscapes dominated by one or a few majority classes. When minority classes are embedded within such regions, their predictions are more susceptible to contextual dominance from surrounding majority classes. By contrast, in more heterogeneous tiles, no single class occupies the local context entirely, and prediction error is correspondingly lower. In this sense, the observed negative correlation suggests that segmentation difficulty is not concentrated in the most fragmented landscapes, but rather in more homogeneous, majority-dominated regions.

4.5.3. Spatial Sieve Analysis: Beyond Boundary Confusion

To directly test whether the prediction difficulty of minority classes is spatially correlated with their proximity to dominant land covers, we conduct a distance-based error analysis. Following the pixel-frequency distribution reported in Section 3.2, we define the three most spatially dominant classes—Dryland (25.93%), Grassland (27.71%), and Shrubland (22.03%), which together account for approximately 75.7% of labeled pixels—as majority classes, and eight classes whose pixel frequencies are more than an order of magnitude below the smallest majority

class (Shrubland, 22.03%), each remaining below 2%, as minority classes: Irrigated Land, Urban Land, Other Construction, Rural Road, Other Transport, Water Bodies, Rural Built-up, and Disturbed Land. Three intermediate classes are excluded, as this analysis focuses on the spatial interaction between the most dominant and the rarest land cover types.

For each test tile, the Euclidean Distance Transform (EDT) is computed on the complement of the majority-class mask, yielding the per-pixel distance $D(x, y)$ to the nearest majority-class pixel. Minority-class pixels are then partitioned into distance zones at 5-pixel intervals (1–5, 6–10, ..., 26–30, and > 30 pixels), and the prediction error rate within each zone is computed as the proportion of misclassified minority-class pixels relative to all minority-class pixels in that zone. Figure 15 presents the resulting error profile across 1,490 test tiles.

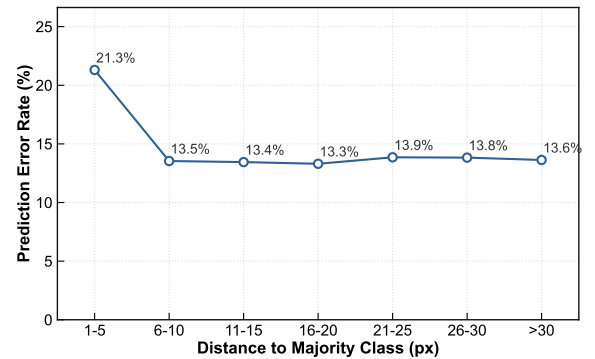


Figure 15: Prediction error rate of minority classes as a function of distance to the nearest majority-class pixel. Error is highest within 1–5 pixels (21.3%) and drops to a stable plateau of 13.3%–13.9% from 6 pixels onwards, with no further reduction beyond 30 pixels (13.6%). This pattern suggests that the effect extends beyond immediate boundary confusion.

Two patterns emerge from this analysis. First, the immediate boundary zone (1–5 pixels) exhibits the highest error rate at 21.3%, indicating that contextual interference is strongest where minority and majority classes are in direct spatial contact. Second, beyond this boundary zone, the error rate drops to a stable plateau of 13.3%–13.9% across all distance intervals from 6 pixels onwards, including beyond 30 pixels (13.6%). This absence of further decrease with distance shows that minority-class pixels remain in an elevated-error regime regardless of their spatial isolation from majority-class boundaries.

This persistent elevation across the entire spatial range indicates that the difficulty models face is not merely local boundary confusion. For geometrically thin minority classes such as Rural Roads and Other Transport, whose widths rarely exceed a few pixels, virtually their entire spatial extent falls within the high-error boundary zone. More importantly, the plateau pattern beyond 6 pixels shows that even spatially extended minority classes—such as Rural Built-up and Water Bodies—do not benefit from interior distance to majority-class boundaries: prediction errors remain around 13.5% across all distance intervals, with no recovery to a baseline level. This distance-based pattern suggests that Spatially Clustered Imbalance operates as a broader context-driven challenge rather than as a purely local boundary phenomenon.

4.5.4. Implications

Taken together, these analyses highlight the role of Spatially Clustered Imbalance in understanding the segmentation difficulty posed by our dataset.

First, the visual and confusion-matrix analyses show that errors on minority classes often manifest as spatial erosion and fragmentation, with misclassifications frequently directed toward spatially adjacent majority classes. Shrub Forest and Dry Cropland, which are both spatially pervasive and spectrally similar to several minority classes, appear as the most frequent misclassification targets.

Second, the landscape-error analysis reveals an inverse relationship between macroscopic landscape complexity and prediction difficulty. Errors are higher in relatively homogeneous, majority-dominated regions, where embedded minority classes are more vulnerable to contextual bias. This pattern can be understood as a macroscopic manifestation of Spatially Clustered Imbalance.

Third, the distance-based analysis indicates that the difficulty is not confined to immediate boundary confusion. Instead, the elevated error persists across the entire interior of minority classes, affecting both geometrically thin classes and more extended ones. This pattern supports the interpretation that Spatially Clustered Imbalance operates as a broader context effect rather than as a purely local boundary phenomenon.

Overall, these findings suggest that Spatially Clustered Imbalance represents a structural challenge arising from the geographic organization of land cover in continuous landscapes. This challenge may not be fully addressed by architecture improvements alone and motivates future research on spatially aware loss design and context-sensitive training strategies.

5. Conclusion

This paper introduces LoessScape, a large-scale semantic segmentation dataset constructed under a continuous-region survey paradigm. By preserving the complete spatial organization of a Loess Plateau region through a semantic-content-aware stratified partitioning strategy, we establish comprehensive benchmarks spanning hierarchical supervised segmentation on the L1 and L2 tasks and unsupervised domain adaptation under spatial distribution shift. Our evaluations show that segmentation difficulty increases markedly under finer semantic granularity and spatial distribution shift, highlighting the challenge of Spatially Clustered Imbalance, in which geometrically thin minority classes remain embedded within strongly autocorrelated dominant terrains.

Beyond aggregate empirical metrics, our cross-disciplinary diagnostic framework provides further insight into model failure in continuous landscapes. Specifically, we identify a counterintuitive Landscape-Error Paradox and use spatial statistical analysis to characterize the phenomenon of Spatial Siege, showing that segmentation difficulty extends beyond localized boundary confusion into broader contextual erosion. Taken together, LoessScape provides a benchmark for studying spatial distribution shift and supports future research on spatially aware loss design, context-sensitive adaptation, and deployment-oriented land-cover mapping.

Data Availability

The LoessScape dataset is publicly available on Zenodo at <https://doi.org/10.5281/zenodo.20340667>, released under the CC BY-NC-SA 4.0 license. A project homepage with full documentation, dataset details, and benchmark results is provided at <https://anyun-hy.github.io/LoessScape/>, and the accompanying repository is hosted at <https://github.com/anyun-hy/LoessScape>.

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